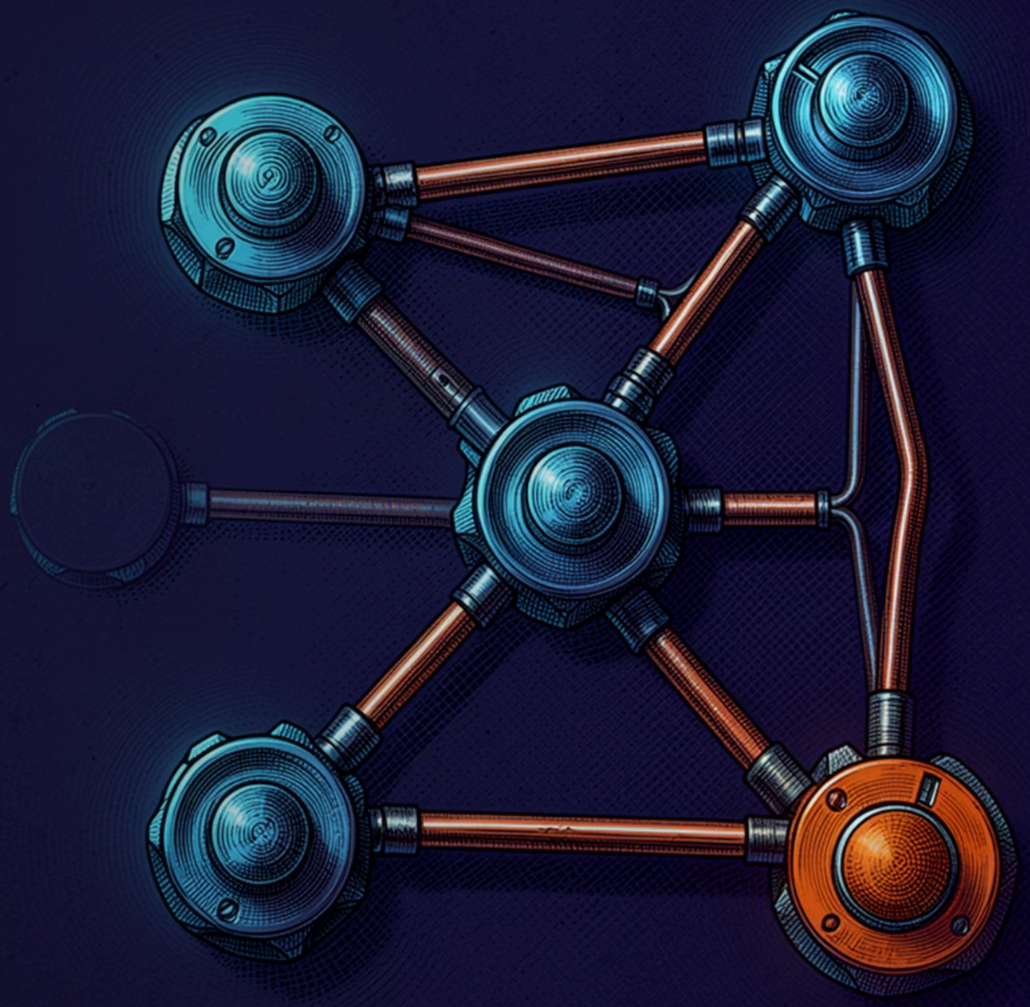
COMPUTER SCIENCE · GRAPH ALGORITHMS

Prim's Algorithm

Finding the Minimum Spanning Tree

Mayank Ninawe • Section D • PRN 25070521182

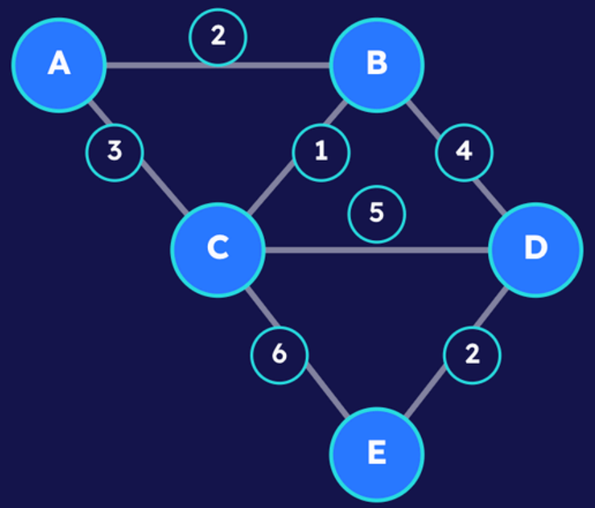


01 · GRAPH BASICS

Graphs turn connections into a structure

- Vertices are nodes; edges connect them.
- Weighted graphs assign each edge a cost or distance.
- A minimum spanning tree connects all vertices at the lowest possible total cost.

Five vertices A–E with weighted undirected edges



Vertices are the points. Edge weights describe the cost of each connection.

02 · THE CORE IDEA

Prim’s Algorithm grows one connected tree

Greedy; for connected, undirected, weighted graphs.

Choose the lightest edge from the tree to an outside vertex.

Useful in network design, roads, wiring, and communication infrastructure.

Stage 1

Start at A; frontier edges from A.

Stage 2

Add the lightest edge to B; check outside edges.

Stage 3

Add the lightest edge to C; update frontier.

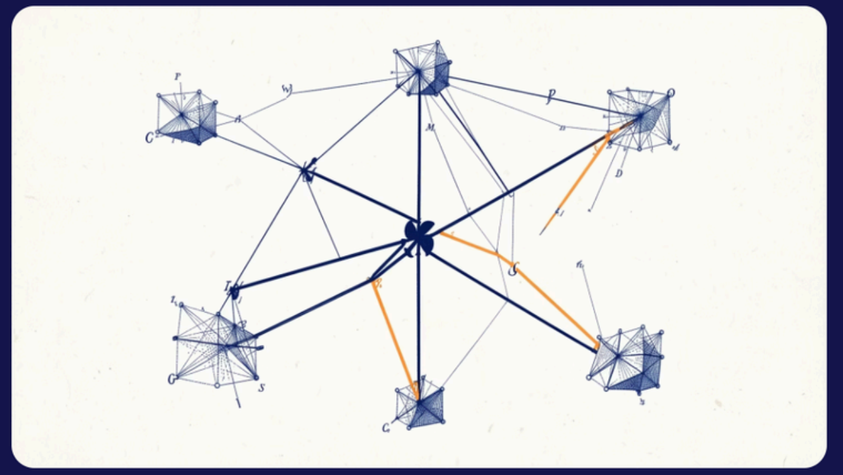
Stage 4

Add the lightest edge to D; show final frontier.

03 · MINIMUM SPANNING TREE

An MST connects all vertices without cycles

A valid MST spans every vertex with exactly V − 1 edges, has the minimum total edge weight, and contains no cycles.



Before selection

All candidate edges are available: A–B=2, A–C=3, B–C=1, B–D=4, C–D=5, C–E=6, D–E=2.

After selection

Selected: A–B=2, B–C=1, B–D=4, D–E=2.

Not selected: A–C=3, C–D=5, C–E=6.

04 · THE ALGORITHM

Each choice expands the tree by one vertex

Start

Start from any vertex.

One vertex is included.

Inspect

Find the minimum-weight edge crossing to an outside vertex.

The frontier supplies candidates.

Add

Add the edge and newly reached vertex.

The tree stays connected.

Repeat

Repeat until every vertex is included.

Stop after V − 1 edges.

05 · WORKED EXAMPLE

Starting at A gives a total weight of 9

Graph edges: A–B=2, A–C=3, B–C=1, B–D=4, C–D=5, C–E=6, D–E=2.

1

Start A → choose A–B=2.

Total: 2

2

Included {A,B} → choose B–C=1.

Total: 3

3

Included {A,B,C} → choose B–D=4.

Total: 7

4

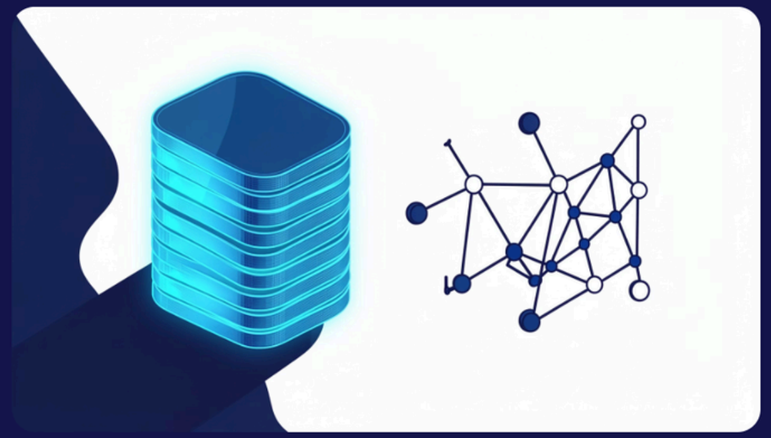
Included {A,B,C,D} → choose D–E=2.

Total: 9

06 · GREEDY IMPLEMENTATION

The greedy rule is simple to implement

A priority queue keeps the next lightest candidate accessible.



Four steps to the MST

1. Choose a starting vertex and mark it included.
2. While some vertices are excluded, find the lightest edge from the tree to an outside vertex.
3. Add the edge and newly reached vertex.
4. Return the MST and its total weight.

07 · PYTHON

Python uses an adjacency list and a min-heap

heapq selects the lightest frontier edge; the visited set skips included vertices.

```
import heapq
def prim(graph, start="A"):
    visited={start}; heap=[(w,start,v) for v,w in graph[start]]; heapq.heapify(heap);
mst=[]; total=0
    while heap and len(visited)<len(graph):
        w,u,v=heapq.heappop(heap)
        if v in visited: continue
        visited.add(v); mst.append((u,v,w)); total+=w
        for n,c in graph[v]:
            if n not in visited: heapq.heappush(heap,(c,v,n))
    return mst,total
graph={"A":[("B",2),("C",3)],"B":[("A",2),("C",1),("D",4)],"C":[("A",3),("B",1),("D",5),
("E",6)],"D":[("B",4),("C",5),("E",2)],"E":[("C",6),("D",2)]}
print(prim(graph)[1])  # 9
```

08 · COMPLEXITY

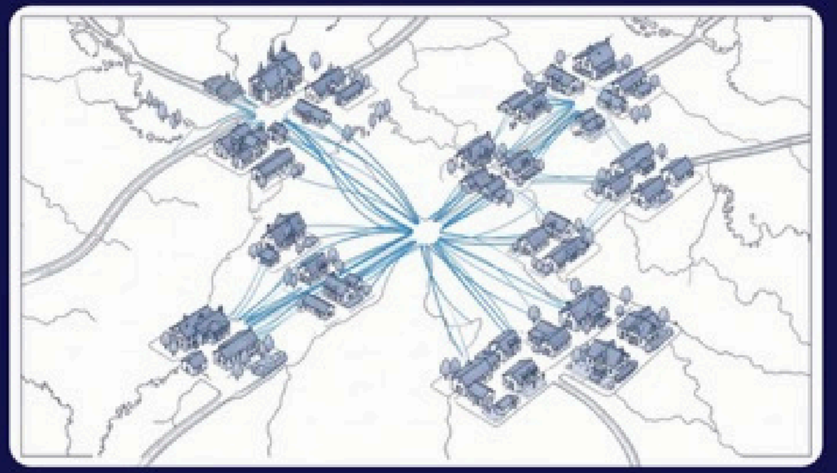
Complexity depends on the graph representation

Representation	Time	Space
Adjacency matrix	$O(V^2)$	$O(V^2)$
Adjacency list + binary min-heap	Typically O(E log V)	O(V + E), plus heap and visited set

V is the number of vertices. E is the number of edges. The adjacency list is usually the practical choice for sparse graphs.

Prim’s Algorithm minimizes connection cost

An MST connects locations at low cost, without redundant cycles.



Designing computer networks

Connecting cities with roads or communication lines

Planning electrical wiring

Building telecommunications infrastructure

Reducing total cost in network design

PRIM'S ALGORITHM

Thank you

Minimum spanning trees connect every vertex at the lowest total cost.

Questions and discussion welcome